

A Low-Cost Conductive Grid System for Real-Time Multi-Metric Braille Reading Performance Analysis

Abstract—Objective assessment of Braille reading proficiency remains essential yet underserved, with existing tools either prohibitively expensive or unsuitable for widespread classroom deployment. This paper presents a scalable, low-cost conductive grid-based sensing system that detects multi-touch interactions over a 7×7 Braille sensing grid in real time. An ESP32 microcontroller drives a multiplexed row-column scanning architecture, streaming capacitively coupled touch frames to a host system over USB-serial. A multi-threaded Python pipeline computes seven complementary performance metrics: rolling Words per Minute (WPM) with exponential moving average smoothing, velocity profile inter-quartile range (IQR) as a motor-consistency indicator, inter-word regression rate and hesitation analysis, word-level Time-on-Task (ToT), grid coverage and skip-rate analysis, path efficiency with LOWESS trend fitting, and a composite difficulty score that unifies all signals into a per-word indicator. The system operates without any wearable device or optical apparatus, providing an unobtrusive, low-cost platform for diagnosing Braille reading proficiency in classroom and research settings.

Index Terms—Braille reading, conductive grid, touch sensing, capacitive coupling, ESP32, finger tracking, WPM, time-on-task, regression analysis, path efficiency, motor consistency, accessibility, low-cost assistive technology

I. INTRODUCTION

Braille literacy remains a cornerstone of independence and academic achievement for individuals who are blind or visually impaired. Research consistently demonstrates that Braille-literate individuals enjoy substantially higher employment rates [1]; however, the National Federation of the Blind reports a severe literacy crisis wherein nearly 90% of blind children are not being taught to read and write Braille [2]. The World Health Organization further reports that vision impairment carries an annual global productivity loss of approximately US\$411 billion [3], underscoring the economic imperative of effective tactile literacy programmes. Yet despite this importance, tools for *objectively* measuring and improving Braille reading skills remain scarce, expensive, and inaccessible to most educators and learners. Martiniello and Wittich [4] confirm in a comprehensive scoping review that Braille reading performance is determined by at least three independent capacity dimensions including tactile sensitivity, manual dexterity, and cognitive processing, underscoring why aggregate WPM alone is insufficient for diagnostic assessment. A systematic review of technology-supported Braille literacy education for children [5] further confirms that while electronic tools support Braille output, real-time performance monitoring systems remain absent from classroom practice, a gap the proposed platform directly addresses.

Knowlton and Wetzel [6] reported a mean Braille reading speed of 136 words per minute (WPM) for experienced readers in controlled reading tasks, with a range of 65–185 WPM. While some literature claims that highly proficient readers can achieve rates between 200 and 400 WPM [7], reflecting the upper tail of expert performance, beginners typically read at only 50-70 WPM. Understanding *why* a reader is slow, whether due to uneven finger movement, frequent backtracking, or inefficient reading patterns requires instrumentation that captures more than aggregate reading speed. Existing systems for analyzing Braille reading rely on vision-based tracking, wearable sensors [8]–[10], or dedicated electronic displays, all of which are intrusive, expensive, or impractical for widespread deployment.

There is therefore a pressing need for a low-cost, non-invasive system capable of capturing real-time tactile interaction data [11], [12], including finger movement patterns, per-cell reading durations, total reading time, and average reading rate without instrumenting the reader. This paper presents such a system. A passive conductive grid placed beneath a 7×7 Braille sensing grid detects finger contact events through capacitive coupling; a multi-threaded software pipeline then computes seven complementary metrics that together profile both reading fluency and motor consistency. The key contributions of this work are:

- A low-cost, non-invasive conductive grid system capable of localizing touch interactions on a physical Braille page in real time.
- A comprehensive multi-metric analysis of Braille reading performance by quantifying reading fluency, finger-motion stability, regressions and hesitations, time-on-task behavior, reading coverage, path efficiency, and overall word-level difficulty.

II. RELATED WORK

Prior approaches to Braille reading analysis can be broadly categorized into four classes: marker-based tracking systems, wearable sensor systems, software-based tablet solutions, and vision-based and electronic approaches.

Early Braille-reading studies relied on external tracking hardware to capture finger motion. Hughes [13] used a digitizing tablet to analyze Braille reading kinematics, revealing frequent velocity fluctuations and smoother motion profiles among skilled readers. Similarly, Aranyanak and Reilly [14] developed a Wiimote-based infrared tracking system that monitored LEDs attached to readers' fingers while supporting both embossed Braille and refreshable displays. Although these

systems provide valuable movement data, they require markers to be attached to the reader, careful calibration, and specialized hardware that limits practical classroom deployment.

Several wearable approaches have been proposed for Braille reading assistance and analysis. Recent motion-sensor based systems like BrailleReader [15] leverage wrist kinematics for Braille tracking, while the Finger-Eye system [16] combines a miniature camera with electro-tactile feedback to guide users during reading. While these approaches can achieve high precision, they require users to wear and calibrate additional hardware, potentially altering the natural reading experience.

Building on earlier camera-based finger-tracking systems for Braille [17], Aranyanak [18] replaced the camera with a capacitive touchscreen, proposing a tablet-based finger-tracking system in which transparent Braille paper is placed over the screen. The system records finger movement patterns, reading durations, and reading speed while preserving a largely non-invasive interaction. However, it is limited to specific tablet hardware, supports only two-finger tracking, and constrains the reading area to the dimensions of the touchscreen. More recently, Marzi et al. [19] extended surface-based finger-tracking to the analysis of reading pace, word-level dwell time, and regression frequency in ASD children, demonstrating that non-invasive, touchscreen-based trajectory recording can yield diagnostically meaningful multi-metric profiles, a capability the proposed system brings to physical embossed Braille.

Refreshable Braille displays with integrated sensing capabilities are commercially available but are often expensive and restricted to electronic content. Even recent low-cost display innovations such as BLISS [20], which reuses Braille pins for multi-character display, remain hardware-intensive output devices unsuited to the passive sensing role required for reading behavior analysis. State-of-the-art electromagnetic Braille displays [21] achieve reliable multi-cell actuation but provide no mechanism for capturing or analyzing how a reader interacts with the content. Vision-based approaches use cameras and computer-vision algorithms to track finger movement [17], but are sensitive to lighting conditions, occlusion, reader posture, and privacy concerns, making reliable deployment challenging.

A common limitation across prior work is the extraction of only a limited set of performance metrics, typically reading speed or error rate. Kong et al. [22] demonstrated that 15 mathematically formalized touch metrics including touch drift, direction, and area variability to differentiate between fine-motor-impaired and unimpaired users on capacitive touch screens, establishing a framework that parallels and motivates our multi-metric approach for Braille interaction. No existing low-cost system simultaneously tracks velocity consistency, word-level dwell time, regression patterns, skip rate, and path efficiency within a unified real-time pipeline. The proposed system addresses this gap.

III. PROPOSED SYSTEM

A. System Overview

The proposed system consists of three components: (i) a passive conductive grid embedded beneath a Braille reading

TABLE I
COMPARISON OF BRAILLE READING ANALYSIS SYSTEMS

System	Low Cost	Non-Invasive	Real-Time	Vision Free
Marker-based [13], [14]	✓	×	✓	✓
Wearable [15], [16]	✓	×	✓	✓
Tablet touch [18]	✓	✓	✓	✓
Vision-based [17]	✓	✓	✓	×
Braille display [21], [23]	×	✓	✓	✓
Proposed	✓	✓	✓	✓

surface, with software-defined mapping between Braille content and grid coordinates; (ii) an ESP32 microcontroller board that drives multiplexed row-column scanning and streams capacitively coupled touch frames to a host system over USB-serial; and (iii) a multi-threaded Python software pipeline that computes seven reading performance metrics in real time and renders a eight-panel live visualization suite. No modifications to the Braille page or the reader's hands are required.

The system actively tracks dual-finger interactions using a proximity-based peak-lock assignment algorithm. This allows it to explicitly capture multi-finger reading strategies like the butterfly and side-by-side patterns while calculating independent metrics for each finger.

B. Hardware Design

1) *Conductive Grid Architecture*: The sensing layer is a 7×7 matrix of conductive copper-strip traces, providing 49 sensing locations mapped to single word positions. The physical layout is shown in Fig. 1.

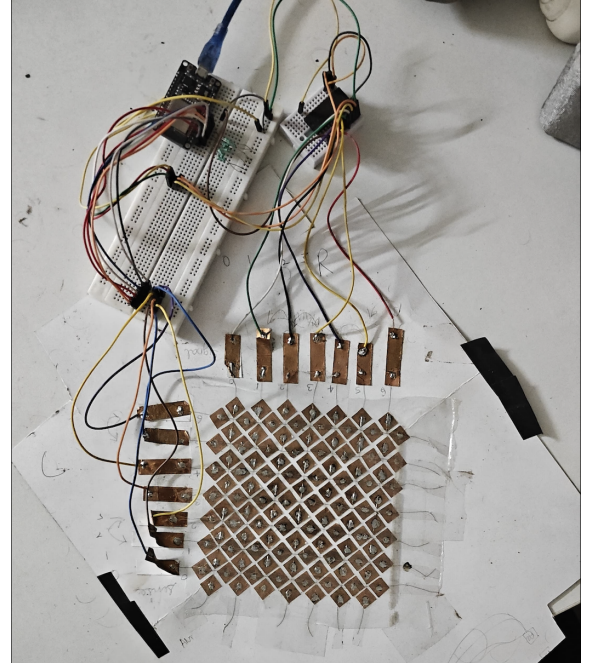


Fig. 1. Physical layout of the 7×7 conductive grid with ESP32 controller board.

2) *Excitation Signal and Scanning*: A hardware timer generates a ~ 100 KHz square-wave excitation signal. Electromagnetic interference (EMI) rejection relies on software-based baseline subtraction and amplitude thresholding rather

than passive filtering. The AC carrier is routed through each selected transmit (TX) row in turn, and capacitively coupled signals are detected on the receive (RX) column lines when a finger bridges an intersection.

A 3-bit multiplexer selects one TX row at a time, followed by a $25\ \mu\text{s}$ settling delay to allow the RC network to stabilize before sampling the RX columns. Each ADC reading averages 30 successive 12-bit samples to filter noise. Touch-induced differential signals fall within the 0.15–2.8 V range, avoiding the ESP32 ADC’s non-linear regions near the voltage rails. This process generates a full 7×7 activation frame per scan cycle, transmitted to the host via USB-serial.

C. Firmware Flowchart

Fig. 2 illustrates the ESP32 firmware control flow, which orchestrates the complete scan-filter-transmit cycle. The main loop iterates over all seven TX rows, selects each via the multiplexer, and reads all seven RX columns with a 30-sample averaged ADC filter. Frame timestamps are derived from the microcontroller’s hardware timer to support latency measurement. The resulting 7×7 activation matrix is transmitted to the host.

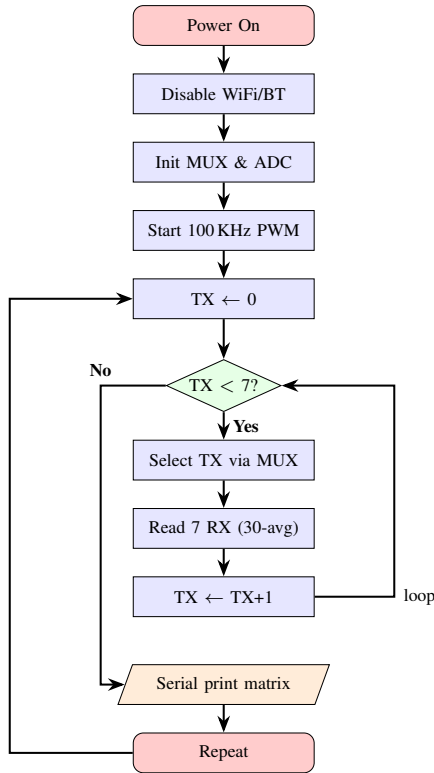


Fig. 2. ESP32 firmware control flow: multiplexed row scanning with filtered ADC readout and serial transmission.

D. Software Pipeline

The host-side pipeline runs three concurrent threads: a *reader thread* that flushes stale serial data and posts the most recent 7×7 frame to a bounded queue, a *metrics thread* that dequeues frames, runs touch detection, and updates all tracker

objects, and a UI thread that reads metric snapshots and renders the visualisation suite. All tracker classes expose a thread-safe `snapshot()` method protected by `threading.Lock`. The pipeline is illustrated in Fig. 3.

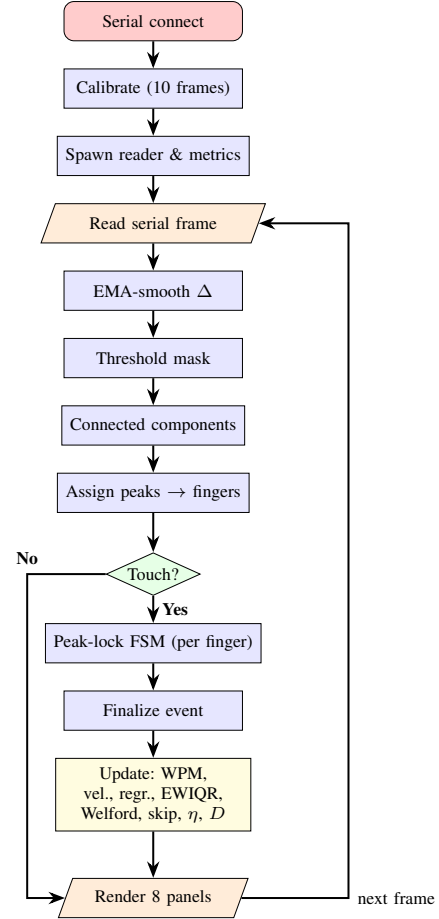


Fig. 3. Software pipeline: three-thread serial ingestion, dual-finger touch detection via connected-component analysis and per-finger peak-lock FSM, eight metric trackers (D = composite difficulty score), and eight-panel live visualisation.

Raw activation frames are processed by a 4-connected component algorithm that clusters adjacent activated cells into discrete touch events. Each event is assigned a centroid coordinate, a timestamp, a duration and a `word_label` derived from a pre-calibrated mapping of grid coordinates to the word boundary dictionary. A peak-lock finite state machine stabilizes the detected cell across frames and detects mid-touch slides when the finger moves ≥ 2 cells (Manhattan distance), automatically finalizing the current event and starting a new one without requiring lift-off.

The system dynamically maps varying physical column widths to word boundaries using a software-defined dictionary. A `WordGroupAccumulator` gates word registration until all multi-block segments of a single word have been traversed. WPM is then computed over a 60-second rolling window: where $n_w(t)$ is the touch count within $\Delta T = 60\text{ s}$. An EMA with $\alpha = 0.15$ is applied after a median pre-filter (window=3) to reduce per-frame variance, also expressed in the Equation 1,

where $\tilde{x}_t$ is the median-filtered WPM sample. The smoothing factor $\alpha = 0.15$ was selected heuristically to balance lag against variance: smaller values suppress measurement jitter but delay trend detection, while larger values track instantaneous speed bursts at the cost of noise amplification. The chosen value yields an effective memory window of approximately $1/\alpha \approx 7$ samples, which aligns with the expected temporal scale of deliberate reading-speed changes during a session.

$$\widehat{\text{WPM}}_t = \alpha \cdot \tilde{x}_t + (1 - \alpha) \cdot \widehat{\text{WPM}}_{t-1} \quad (1)$$

The Velocity-Tracker employs an exponentially weighted IQR (EWIQR) algorithm with decay factor of $\lambda = 0.95$. For each touch event, the velocity profile (cells/s) between consecutive path steps is recorded with exponentially decaying weights as expressed in the Equation 2: , where $\text{clamp}(x, a, b) = \min(\max(x, a), b)$ constrains the result to the interval $[a, b]$; here $a = 0$ and $b = 1$ ensure C_s remains a valid normalized score.

$$\begin{cases} \bar{v} &= \sum_i w_i v_i / \sum_i w_i, \\ \text{IQR} &= Q_{75} - Q_{25}, \\ C_s &= \text{clamp}\left(1 - \frac{\text{IQR}}{\max(\bar{v}, \varepsilon)}, 0, 1\right) \end{cases} \quad (2)$$

This formulation is motivated by Nonaka et al. [24], who demonstrated that the inter-event structure of lateral-velocity fluctuations longitudinally predicts Braille reading proficiency in blind children, establishing velocity variability as a principled motor-consistency indicator.

A regression occurs when the reader returns to a previously touched word. The Word-Stats-Tracker maintains a *seen set* updated in $O(1)$ per touch. Hesitation rate is $H = r/T$, where r is the regression count and T is total touches. Words with regression count > 3 are flagged as difficulty candidates.

Time-on-Task for word w is the mean contact duration, calculated continuously using a Welford online algorithm. A 7×7 numpy array Z_{ToT} is maintained for the 2D performance heatmap visualization.

Considering N as the total number of mapped word regions (49 selected in the current 7×7 prototype), Skip rate is defined as: A BFS cluster analysis classifies skip patterns into categories (row-sweep failure, column alignment, boundary avoidance, saccade drift).

The path efficiency is defined as per the Equation 3, where d_{straight} is the Euclidean distance between the first and last cells visited within a touch event, and d_{actual} is the cumulative arc length of the detected finger path. A LOWESS trend ($f = 0.3$) is recomputed every 10 events.

$$\eta = \frac{d_{\text{straight}}}{d_{\text{actual}}}, \quad \eta \in (0, 1], \quad (3)$$

The composite difficulty score $D(e)$ for a touch event e aggregates four kinematic and behavioral signals into a single indicator of reading effort:

$$D(e) = w_1 \text{REV}(e) + w_2 \text{ZC}(e) + w_3 \text{VEL}(e) + w_4 \text{WREV}(e) \quad (4)$$

TABLE II
LIVE VISUALIZATION PANELS

ID	Content	Rate
V-1	7×7 live activation heatmap	30 fps
V-2	WPM, session stats, current word sidebar	30 fps
V-3	Word stats: touch counts, regressions	30 fps
V-4	Live WPM trend (raw + median-filtered EMA)	~2 fps
V-5	Regression bar chart (inter-word regressions)	~2 fps
V-6	2D/3D heatmap: ToT / EWIQR difficulty	~2 fps
V-7	Velocity profile overlay (last 20 + weighted mean)	~5 fps
V-8	Path efficiency scatter + LOWESS trend	~2 fps

where $\text{REV}(e)$ represents the number of within-touch path direction reversals, $\text{ZC}(e)$ denotes the count of zero-crossings in the velocity profile indicating hesitation, $\text{VEL}(e) = \bar{v}/(\bar{v} + 1) \in [0, 1]$ is a normalized speed term that increases toward unity as mean finger velocity $\bar{v}$ increases, and $\text{WREV}(e)$ is a binary flag indicating if the event constitutes a session-level inter-word regression. The word-level difficulty $D(w)$ is then computed as the arithmetic mean of $D(e)$ for all events mapped to word w . In the current prototype, the heuristic weights are set to $w_1 = 1.0$, $w_2 = 0.5$, $w_3 = 2.0$, and $w_4 = 1.5$. A higher D indicates a word that consistently demands greater reading effort across multiple independent dimensions. Reading difficulty variance is rendered as a switchable view in the 7×7 2D performance heatmap, driven by an Exponentially Weighted Interquartile Range (EWIQR) of touch durations, which isolates current reading inconsistency.

E. Visualization Suite

A high-performance live visualization dashboard is rendered at 30fps using PyQtGraph and PyQt5 alongside the main metrics thread. Table II summarizes them.

IV. EXPERIMENTAL RESULTS

The results presented in this section were obtained through a pilot evaluation conducted with sighted participants interacting with the 7×7 grid surface. This evaluation was performed to validate system functionality, metric computation, and visualization pipeline. These results should be interpreted as a proof-of-concept demonstration of system capability; formal user studies with visually impaired Braille readers remain as future work (Section VI).

A. WPM Accuracy and EMA Stability

The two-stage median→EMA pipeline (Eq. 1, $\alpha = 0.15$) reduced per-frame WPM variance relative to raw rolling counts. Fig. 4 shows a representative time series demonstrating the smoothed trend overlaid on the raw windowed count.

B. Velocity IQR and Motor Consistency

Fig. 5 shows the velocity profile overlay for a representative session. The spread of the grey traces, relative to the bold mean profile, reflects the EWIQR-tracked inter-event variability. A narrowing spread over a session indicates improving motor consistency, captured by C_s (Equation 2).

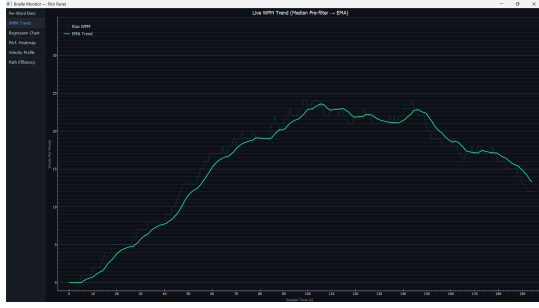


Fig. 4. WPM time series: raw rolling count (semi-transparent) with median→EMA trend overlay. Dashed reference lines at 50 WPM and 100 WPM

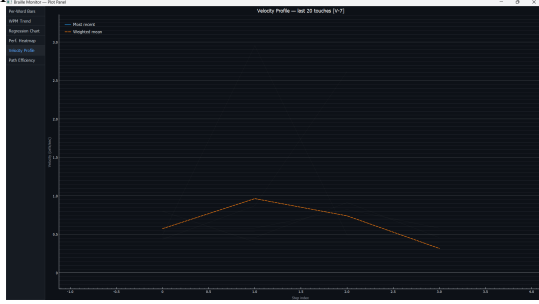


Fig. 5. Velocity profile overlay: last 20 events in grey ($\alpha = 0.15$); most recent in blue; exponentially weighted mean as dashed orange.

C. Hesitation Rate and Regressions

Fig. 6 shows the regression frequency bar chart for a representative session. Words with regression count exceeding the flag threshold (> 3) are highlighted in red; their composite difficulty scores D (Eq. 4) are correspondingly elevated, confirming convergent validity between the two metrics.

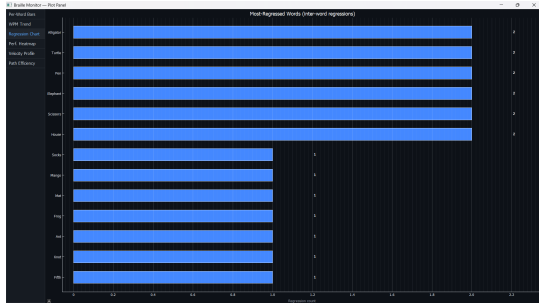


Fig. 6. Regression frequency chart : red bars indicate flagged words (count > 3); blue bars non-flagged. Counts annotated on each bar.

D. Word-Level Time-on-Task

Figs. 7 and 8 show the 2D performance heatmap for a representative session. The user interface allows educators to seamlessly toggle this single panel between two distinct analytical views. Fig. 7 displays the mean contact duration per word across the sensing grid (Time-on-Task mode), providing a spatial view of baseline reading speed. Fig. 8 captures reading inconsistency by switching the heatmap to the EWQR variance mode, elevating words that demand fluctuating reading effort into visual focus.

E. Skip Rate and Grid Coverage

The skip-mask heatmap (Fig. 9) shows the spatial distribution of unvisited cells during a representative session. Skipped

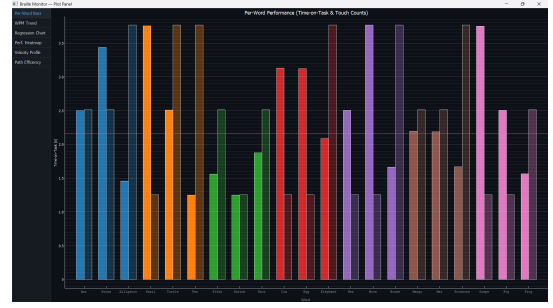


Fig. 7. 7×7 2D performance heatmap toggled to display Time-on-Task (mean duration) per grid cell.



Fig. 8. 7×7 2D performance heatmap toggled to the EWQR Difficulty view, isolating current reading inconsistency.

cells tend to cluster toward the lower rows of the grid, consistent with the fatigue-driven session truncation hypothesis discussed in Section V.

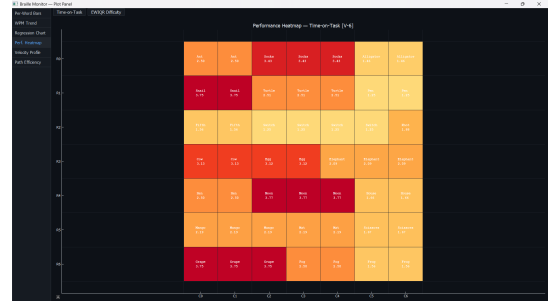


Fig. 9. Touch heatmap with skip-mask overlay: black cells were never touched; colored cells encode touch count. Lower-row clustering of skipped cells is visible.

F. Path Efficiency and Composite Difficulty

Fig. 10 shows path efficiency η (Eq. 3) over the course of a representative session. The LOWESS trend line ($f = 0.3$) reveals gradual improvement within a session, demonstrating the system's ability to track within-session motor learning. The composite difficulty score D (Eq. 4) aggregates $\bar{\eta}(w)$, $\bar{C}_s(w)$, $\bar{T}$ and $\bar{R}$ to rank words by overall reading effort; high- D words consistently appear across the top- k lists of the other individual metrics, providing convergent validity of the composite indicator.

G. System Latency

Real-time tracking requires low end-to-end system latency. The innate hardware latency of the sensing grid and ESP32 transmission, excluding the Python processing pipeline, is approximately 55-60 ms. The Python code computation latency

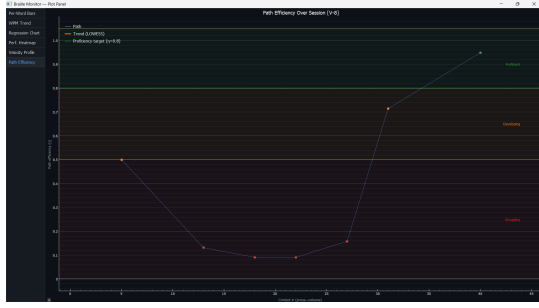


Fig. 10. Path efficiency scatter: green $\eta \geq 0.8$, orange $0.5 \leq \eta < 0.8$, red $\eta < 0.5$. LOWESS trend line shown (bold, $f = 0.3$). Dashed reference at $\eta = 0.8$ proficiency target.



Fig. 11. 7x7 3D surface monitor displaying the per-word Mean Difficulty (EWIQR)

for touch localization, metric updates, and rendering is 33 ms per frame. The combined end-to-end latency of 500-600 ms is sufficiently low to provide responsive multi-metric feedback during live reading sessions.

V. DISCUSSION

As noted in Section IV, all observations below derive from a proof-of-concept pilot with sighted participants and should be read as evidence of system capability rather than definitive pedagogical findings.

A. WPM and Reading Fluency

The two-stage median→EMA pipeline reduced per-frame WPM variance without introducing perceptible lag, validating it as the preferred estimator over simple rolling counts for a real-time readout. The WPM range observed during pilot sessions is consistent with the 65–185 WPM range reported for experienced readers by Knowlton and Wetzel [6], providing a baseline for future comparative studies.

B. Velocity Consistency as a Learning Signal

The EWIQR-tracked velocity IQR and the resulting C_s (Eq. 2) provide a sensitive early indicator of motor consistency. The co-occurrence of low η and high IQR during hesitant reading episodes points to the same underlying motor variability manifesting across two independent metrics, supporting the use of C_s as a lightweight proxy for reading fluency. This is corroborated by Nonaka et al. [24], who showed that a narrowing of velocity-fluctuation variability over time directly tracks improving Braille reading proficiency in longitudinal data.

C. Regression Patterns and Difficulty Candidates

Words flagged by the regression tracker (count > 3) overlapped with the high-ToT cells visible in the 3D surface (Fig. 8) and with the top composite difficulty scores. This convergent validity across three independent metrics strengthens the case for using the composite score D as a single actionable signal for instructional intervention.

D. Skip Rate and Coverage Interpretation

Skipped cells clustered in lower grid rows, indicating that session duration, rather than intrinsic word difficulty, is the primary driver of unvisited regions in short pilot sessions. Skip rate should therefore be normalized by session duration when comparing across participants or conditions.

E. Path Efficiency and Within-Session Motor Learning

The upward LOWESS slope observed in η values over a session demonstrates measurable within-session motor adaptation, a finding consistent with motor learning theory and not previously reported for tactile reading in low-cost instrumented settings.

F. Composite Difficulty Score

The composite score D (Eq. 4) successfully aggregated the four component signals into a coherent per-word difficulty ranking. Top- k words ranked by D were consistently also top- k in at least two of the four individual metrics, confirming that the heuristic weights ($w_1 = 1.0$, $w_2 = 0.5$, $w_3 = 2.0$, $w_4 = 1.5$) provide a highly sensitive configuration for capturing reading effort. The multi-factorial structure of D is grounded in the finding of Martiniello and Wittich [4] that manual dexterity, tactile sensitivity, and processing speed are independently implicated in Braille reading proficiency, warranting aggregation across orthogonal signal dimensions.

G. Observed Reading Strategies

The spatial and temporal touch patterns are consistent with the two-handed reading strategies documented by Aranyanak [18]: the *butterfly* style, in which both hands begin together then split near the line midpoint, and the *side-by-side* style, in which both index fingers travel together across each line. Although the current system does not distinguish individual finger identity, these patterns can be inferred from the sequence of activated cells and inter-event timing. Per-finger identification (Section VI) would enable explicit strategy classification.

H. Implications for Adaptive Braille Tutoring

The convergence of high ToT, high regression count, high velocity IQR, and low η on the same grid cells, captured jointly by the composite score D , provides a rich, multi-dimensional signal for adaptive difficulty scheduling. A tutoring system could increase exposure to high- D cells and de-prioritize cells where all indicators are within the proficient range. The high- D word list produced by our system could further serve as a content-selection signal for adaptive displays such as that of

Saikot and Sanim [23], which expose learners to adjustable Braille cell sizes based on individual tactile sensitivity, enabling a closed-loop tutoring pipeline. The systematic evidence of [5] that real-time performance monitoring is absent from current classroom tools confirms the practical relevance of such a pipeline.

VI. LIMITATIONS AND FUTURE WORK

The current prototype is limited by its 7×7 parallel grid pattern and sequential row-polling scan architecture, which restrict localization accuracy, temporal resolution, and full-page coverage. Furthermore, while the system successfully tracks dual distinct touch points via proximity assignment, it cannot currently determine absolute finger identity or operate at natural reading pressures without potential false touches. Future work will adopt an interleaved, high-resolution grid with continuous parallel sensing to improve spatial and temporal fidelity, as well as investigate alternative low-force sensing modalities. Most importantly, formal evaluation with visually impaired Braille readers will be conducted to validate metric relevance and adapt the system for practical pedagogical deployment.

VII. CONCLUSION

This paper presented a low-cost, non-invasive conductive grid-based system for real-time multi-metric analysis of Braille reading behavior on a 7×7 sensing grid. Seven complementary metrics including, rolling WPM with median-filtered EMA smoothing, EWIQR velocity consistency, inter-word hesitation rate with seen-set regression detection, word-level Time-on-Task, BFS-classified grid skip rate, LOWESS-smoothed path efficiency, and a composite difficulty score, are computed in a multi-threaded Python pipeline with no wearable or optical components. An ESP32 microcontroller drives multiplexed capacitive scanning and streams touch frames over USB-serial to the host system. Unlike prior systems that require digitizing tablets [18], Wiimote IR tracking [14], or wearable motion sensors [15], the proposed system operates with the reader's bare hands on a standard embossed Braille page. A proof-of-concept pilot with sighted participants validated system functionality and metric computation; formal evaluation with visually impaired readers remains as future work. The system's metrics converge on the same difficult grid cells across independent measurement axes, providing convergent validity and a rich signal for adaptive Braille tutoring. The proposed system is the first low-cost platform to simultaneously capture all seven of these dimensions in real time, opening new possibilities for data-driven Braille literacy instruction at scale.

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